DAYANANDA SAGAR UNIVERSITY
SCHOOL OF ENGINEERING

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REPORT FILE OF MINOR PROJECT PHASE – I

VoltRescue – Power When You Need It

Submitted By

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(2024-2025)

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CERTIFICATE

This is to certify that the work titled "**VoltRescue** – Power When You Need It" is carried out by K V Sai Sudheendra kumar-ENG22AM0105, K M N S Silesh-ENG22AM0100 , G Deveswar Reddy-ENG22AM0090,K Teja Sri SaiChakradhar-ENG22AM0106 and L Harshavardhan Reddy-ENG22AM0111 Bonafide students of Bachelor of Technology in Computer Science and Technology at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Technology, during the year 2024-2025.

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DECLARATION

We, K V Sai Sudheendra kumar-ENG22AM0105, K M N S Silesh-ENG22AM0100 , G Deveswar Reddy-ENG22AM0090,K Teja Sri SaiChakradhar-ENG22AM0106 and L Harshavardhan Reddy-ENG22AM0111, are students of the sixth semester B.Tech in Computer Science and Technology, at School of Engineering, Dayananda Sagar University, hereby declare that the project phase - I titled "Project Title" has been carried out by us and submitted in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Technology during the academic year 2023-2024.

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ABSTRACT

The growing demand for electric vehicles (EVs) is transforming the transport industry, but range anxiety and a lack of charging infrastructure are still major concerns for EV users. VoltRescue provides a fresh solution by providing on-demand mobile EV charging, meaning drivers never lose power. Customers can merely place emergency charging orders through a simple app, and a VoltRescue unit will be sent to their precise location, providing quick and convenient power.

VoltRescue combines advanced technology, such as GPS tracking, real-time request processing, and fast-charging units, to offer a simple experience. In contrast to traditional charging stations with diversions and long queues, VoltRescue delivers power to the car, minimizing time and maximizing convenience. VoltRescue is particularly convenient in city centers, highways, and rural areas where charging stations are scarce.

With the worldwide EV market continuing to expand, VoltRescue is a very lucrative business opportunity with tremendous growth potential. Operating on a pay-per-use business model with room for subscription-based products and fleet collaborations, the business has a scalable and lucrative business model. By filling the EV infrastructure gap and offering mobile charging services with uptime guarantees, VoltRescue not only enhances the convenience of EV owners but also accelerates the transition to a greener and electric-powered world

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1 Introduction

The use of electric vehicles (EVs) is increasing at a fast pace as the globe is transforming towards eco-friendly transport. One of the biggest problems EV user's encounters is range anxiety—the apprehension of draining the battery with no charging point nearby. Poor charging infrastructure, particularly on highways and rural regions, also contributes to this problem. VoltRescue plans to solve this problem by offering an on-demand mobile EV charging service, which will make drivers never stranded with a low battery.

VoltRescue works through an easy-to-use app where customers can order emergency charging help. A local VoltRescue vehicle with high-speed charging units is sent out to them, offering quick and convenient charging help. Through the use of GPS, live request management, and effective power transfer, VoltRescue cuts down on the hassle of looking for charging stations, making EV ownership more convenient and hassle-free.

As EVs gain wider adoption, VoltRescue has a huge potential to fill the gap in charging infrastructure. The platform is scalable and can serve individual EV owners as well as fleet operators. By providing a pay-per-use model combined with subscription-based services, VoltRescue makes it accessible, convenient, and reliable. This project not only improves the EV user experience but also drives the transition to a cleaner, greener future.

2 LITERATURE REVIEW

Challenges in EV Charging Infrastructure:

Research shows that traditional EV charging stations are expensive, require a lot of space, and take time to wait (Neaimeh et al., 2017). Most of the stations are in the city center, highways have a minimal number of charging stations, and rural areas have almost none. This lack of availability discourages people from using EVs, fearing that they will run out of battery and lack a place to charge (Zhang & Wang).

Emerging Solutions: Mobile EV Charging

To overcome these gaps, researchers have examined on-demand and mobile EV charging as an efficient and scalable option. Mobile charging points deliver power wherever it is required, minimizing the need for fixed locations. Improvements in battery tech, GPS monitoring, and real-time demand control have enabled increased mobile EV charging feasibility (Chen et al., 2020). Research indicates that incorporation of AI-based fleet management can further refine service delivery to meet quick response times to stranded drivers (Smith et al., 2023).

VoltRescue's Innovation and Market Position

Existing EV charging networks like Tesla Superchargers, ChargePoint, and Ionity focus on fixed stations, requiring heavy infrastructure investment and limited coverage. VoltRescue offers a flexible, mobile, on-demand charging service, using GPS tracking and real-time request management to reach drivers anywhere. With high-speed charging and a pay-per-use or subscription model, VoltRescue makes EV charging more accessible, affordable, and convenient.

Conclusion

Literature supports that mobile EV charging is an essential innovation in overcoming the restrictions of conventional charging infrastructure. Through offering a go-anywhere charging option, VoltRescue boosts the dependability of EVs and facilitates quicker take-up of green mobility. As worldwide demand for EVs keeps on increasing, VoltRescue stands to be a top contender in filling the EV infrastructure gap, making electric mobility more convenient.

3 BUSINESS MODEL

3.1 Idea and Competition

Business Idea:

The exponential rise of electric vehicles (EVs) has generated huge demand for cost-effective and efficient charging systems. Yet, the current fixed charging system is inadequate, and this has resulted in range anxiety and discomfort among EV owners. VoltRescue is a revolutionary on-demand mobile EV charging service aimed at overcoming these limitations. With an easy-to-use mobile application, EV drivers can order emergency charging services at any time and location. A VoltRescue vehicle with advanced high-speed charging technology is sent to the customer's location for instant and guaranteed service. This avoids detours and waiting in line at fixed charging stations, making EV ownership convenient and hassle-free.

VoltRescue employs real-time GPS tracking, AI-based demand management, and dispatch-efficient architecture to maximize service delivery. With a mobile solution, VoltRescue offers charging convenience to the highway, rural, and under-served urban locations where fixed stations are impractical. The service is pay-per-use with add-on subscription plans for heavy users and fleet operators.

Market Competition:

The EV charging sector is currently dominated by fixed network charging station businesses, such as Tesla Superchargers, ChargePoint, Ionity, and Electrify America. They are expensive, space-intensive, and time-consuming to install but offer required infrastructure. They are largely located in urban areas, and highways and rural regions are still under-developed.

VoltRescue offers a mobile, on-demand EV charging solution, eliminating the need for expensive fixed stations. It quickly reaches stranded drivers using AI-based fleet management and real-time tracking. Serving both individual EV owners and fleets, VoltRescue makes EV adoption easier and more convenient.

Through strategic partnerships with EV manufacturers, ride-hailing companies, and roadside assistance companies, VoltRescue is poised to own the market for mobile EV charging with a game-changing and scalable solution that renders fixed charging networks redundant.

3.2 Customer and Need

VoltRescue is designed for electric vehicle (EV) owners who are concerned about running out of battery while driving. Our target consumers are city commuters, long-distance drivers, fleet operators, and ride-hailing drivers who use EVs for daily activities. Our target consumers are typically between the ages of 25 and 55, tech-literate, environmentally conscious, and willing to pay for EV technology. They seek convenient solutions that are environmentally friendly and assist with mobility concerns.

In order to effectively reach our customers, VoltRescue will utilize several different marketing channels. Social media marketing on Instagram, Facebook, and LinkedIn will raise awareness and engage potential users. Blog posts, informing users about our service. Google and social media paid ads will capture users looking for EV charging solutions. Furthermore, partnering with EV manufacturers, ride-sharing, and fleet operators will enable us to reach even more people. Marketing our mobile application, providing referral rewards, and partnering with local roadside assistance services will further boost our customer acquisition plan.

3.3 Financials and Revenue Model

VoltRescue needs initial capital to buy trucks, mobile EV charging machines, develop an app, and cover operating costs and marketing. The initial expenses will include setting up a fleet, developing software, salaries for employees, and licensing fees. On top of that, operating costs like maintenance, fuel, and customer support will be included, as well as a separate marketing budget for internet ads, collaborations, and promotional events.

We have different ways to make money. Customers will pay based on how long they charge their cars and how much energy they use. We will also have a subscription option for regular users, including fleet operators and ride-hailing companies, and give them faster service and discounts. We will also make more money with partnerships with electric vehicle manufacturers, ride-hailing companies, and roadside assistance companies. Additionally, we will make more money with ads in our app and premium features like priority bookings and AI route planning.

We expect to break even within 18 to 24 months, subject to customer take-up and market penetration. With increasing take-up of EVs and demand for mobile charging technology, profitability in three years is expected based on scalable business, increasing customer base, and strategic partnerships in the EV value chain.

4 Software Requirements

Tech Stack:

VoltRescue needs a scalable and strong tech stack to support seamless real-time operation, easy-to-use interfaces, and efficient backend management. The technologies selected are as follows:

Frontend: React Native for cross-platform mobile app development, enabling effortless deployment on Android and iOS.

Backend: Node.js with Express.js for efficiently managing API requests.

Database: Firebase Fire store to facilitate real-time database management in order to synchronize data easily across customers and service providers.

Authentication: Firebase Authentication for secure, simple user login and management.

Mapping & Location Services: Google Maps API and Geolocation API for real-time location tracking, routing, and service provider dispatch.

Payment Gateway: Stripe or Razorpay for safe and easy in-app transactions.

Cloud Storage: Firebase Storage to hold user-uploaded content such as images or documents.

Push Notifications: Firebase Cloud Messaging (FCM) for delivering real-time updates and notifications to users.

Hosting & Live Demo:

For reliable performance and scalability, VoltRescue will be hosted on cloud-based platforms:

Backend Hosting: Google Cloud Platform (GCP) or AWS Lambda for scalable serverless computing, reducing operational costs while ensuring high availability.

Database Hosting: Firebase Firestore, offering real-time synchronization and low-latency data access.

Frontend Deployment: Google Play Store and Apple App Store for mobile app distribution.

Web Hosting (if applicable): Vercel or Netlify for hosting the website, ensuring fast load times and seamless deployment.

Live Demo Platform: A dedicated staging environment on Firebase Hosting or AWS Amplify for testing and demonstrations before full deployment.

This tech stack and hosting strategy ensure VoltRescue operates smoothly, providing a reliable and scalable EV charging assistance platform for users.

References

A Appendix

A.1 Logo Design Report [Maximum of 300 words]

Figure 1: VoltRescue – Power When You Need It

The VoltRescue logo represents visually the company's vision of offering fast, dependable, and on-demand EV charging solutions. The central figure is a streamlined electric car, representing new-age transportation and green mobility. The lightning bolt indicates immediate power delivery, reinforcing the company's primary value—efficiency of roadside EV charging. The circular shape with a global grid emphasizes the brand's vision of accessibility, implying a global network that seeks to assist electric vehicle drivers anywhere, anywhere. The charging plug indicates the brand's dedication to clean energy and technological innovation, making it immediately recognizable in the EV space.

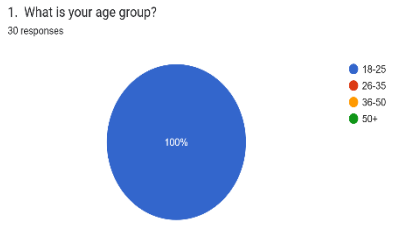
The color choices are picked to show what the brand is about. The main blue color shows trust, reliability, and technology, making customers feel confident in VoltRescue's dependability. The green accents stand for sustainability and being good for the environment, highlighting the company's part in lowering carbon emissions by helping people use electric vehicles. The white highlights make things clear and provide contrast, giving the logo a clean and modern look that connects with innovation and progress.

The bold, modern font is easy to read and shows confidence. The uppercase letters create a sense of urgency, matching the company's fast response promise. The tagline, "Power When You Need It," highlights VoltRescue's commitment to helping EV drivers. Overall, the logo effectively represents the brand's mission and vision

A.2 Survey Report

The information presented in this report has been collected through a Google Form survey, where participants shared their insights on EV charging challenges and mobile charging solutions. The survey helped gather valuable data on user preferences, demand for on-demand EV charging, and potential concerns. Responses were analyzed to understand market needs and improve the VoltRescue service. This feedback plays a crucial role in shaping our business strategy and service offerings.

https://docs.google.com/forms/d/e/1FAIpQLSfua5ql1fZLyr_D2ocQkHhmGzxVTe4OdS7D0xArIfon5iA7vg/viewform?usp=dialog



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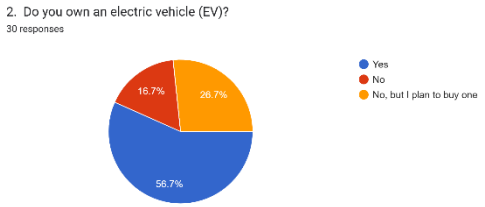

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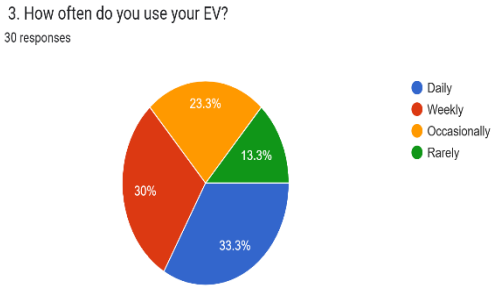

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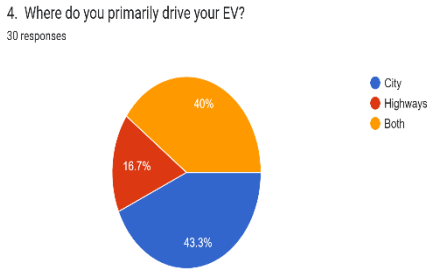

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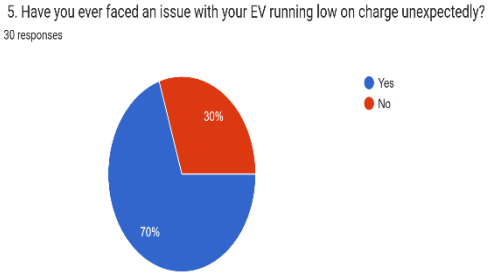

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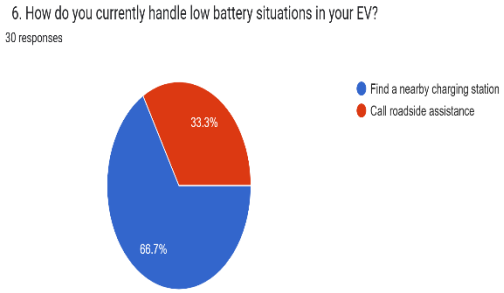

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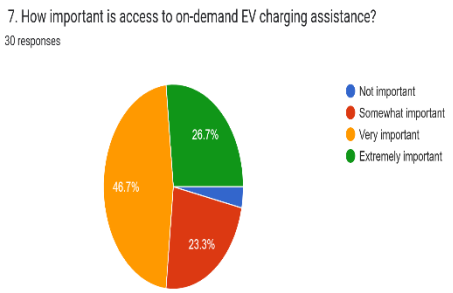

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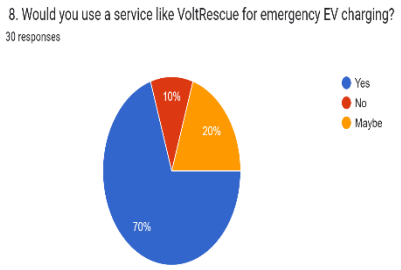

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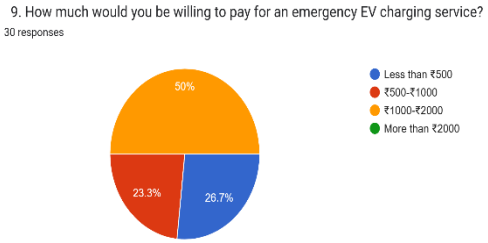

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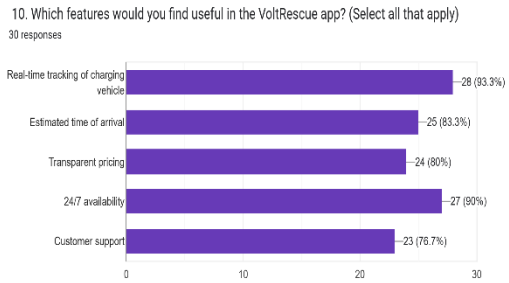

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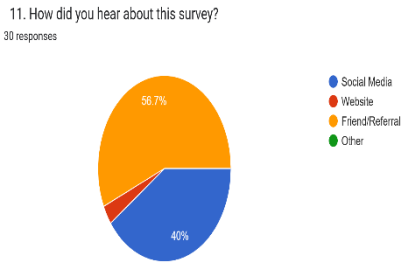

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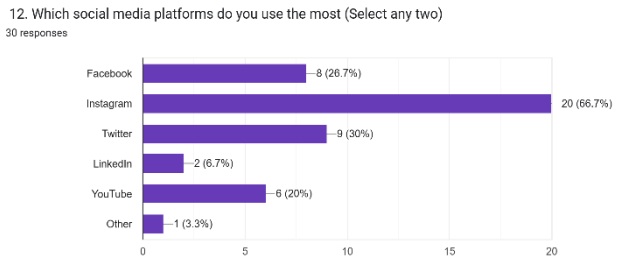

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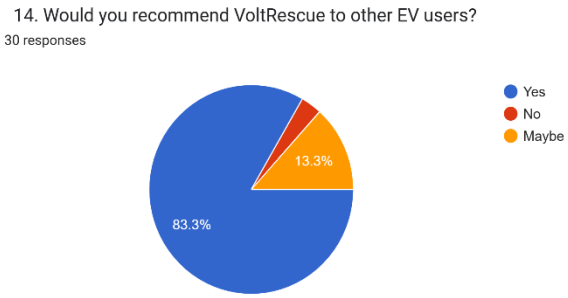

Figure 14

A.3 Interview Report

This section has more details.

A.4 Data collection/Abstraction Report